FLORIDA ATLANTIC UNIVERSITY

ISM 4403-100 11453

Advanced Business Analytics

3 Credit(s)

Spring 2026 - 1 Full Term

Instructor Information

Mohsen Emadikhiav
Email: memadikhiav@fau.edu
Office: Barry Kaye Hall (KH 150)
Office Hours: By Appointment (Online: In Canvas you can navigate through “Zoom > Appointments” to book a meeting.)

Course Description

Advanced Business Analytics
Prerequisite: ISM 3116
An in-depth examination of business analytics methods of visualization, data mining, text mining and web mining using various analytical tools. Applications to smaller firms are investigated in a laboratory setting.

Prerequisites/Corequisites

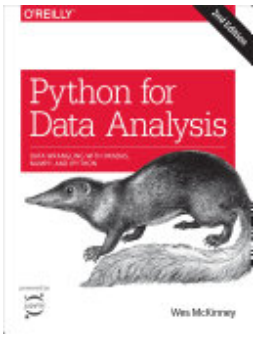
Prerequisite(s): The following course:

- ISM 3116 Graduate / Undergraduate (Minimum Grade of C)

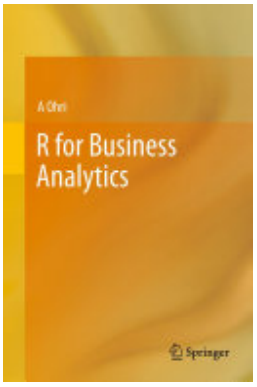
Instructional Method

Fully Online Class
100% of the course is delivered online.

Required Texts/Materials



(Not Required) Python for Data Analysis
ISBN: 9781491957615
Authors: Wes McKinney
Publisher: "O'Reilly Media, Inc."
Publication Date: 2017-09-25



(Not Required) R for Business Analytics
ISBN: 9781461443438
Authors: A Ohri
Publisher: Springer Science & Business Media
Publication Date: 2012-09-14

Course Objectives/Student Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Understand, write, and run Python codes. (CO: 1)
2. Scrape data from websites in different formats with Python. (CO: 2)

3. Store your data in databases and retrieve it with Python. (CO: 3)
4. Cleaning and manipulating data with Python. (CO: 4)
5. Visualize data with Python (Matplotlib library). (CO: 5)
6. Use different libraries in Python, such as Pandas, to analyze data. (CO: 6)
7. Understand, write, and run R codes. (CO: 7)
8. Cleaning and manipulating data with R. (CO: 8)
9. Visualize data with R (using ggplot2). (CO: 9)
10. Use available packages in R to analyze data. (CO: 10)

(CO= Course Level Objective)

Faculty Rights and Responsibilities

Florida Atlantic University respects the rights of instructors to teach and students to learn. Maintenance of these rights requires classroom conditions that do not impede their exercise. To ensure these rights, faculty members have the prerogative to:

• Establish and implement academic standards.

• Establish and enforce reasonable behavior standards in each class.

• Recommend disciplinary action for students whose behavior may be judged as disruptive under the Student Code of Conduct University Regulation 4.007.

Disability Policy

In compliance with the Americans with Disabilities Act Amendments Act (ADAAA), students who require reasonable accommodations due to a disability to properly execute coursework must register with Student Accessibility Services (SAS) and follow all SAS procedures. SAS has offices across three of FAU's campuses – Boca Raton, Davie and Jupiter – however disability services are available for students on all campuses. For more information, please visit the SAS website at www.fau.edu/sas/.

Course Evaluation Method

Assignments (50% total course grade)

- 7-10 Assignments, up to 100 points each

You will apply the skills and concepts covered in the module. Detailed instructions, helpful resources, and grading rubrics are provided in the Canvas course. Assignment pages will close at the stated due date and time. **Late submissions will not be accepted.**

Discussion (10% total course grade)

For each module, students should add at least one post under the discussion of that module. The post can be an exciting idea, application of the materials, assignment, general opinion, etc. Students also should comment under at least two posts from other students. **Late submissions for discussions will not be accepted.**

Final Project (40% total course grade)

You will be required to participate in a team project that will involve collaborating with other students to work on a case study that involves data cleaning using Python, performing preliminary analysis on the data, and finally reporting your findings and recommendations. More details will be provided during the semester.

This assignment offers you the opportunity to practice virtual collaboration skills that are applicable to the 21st Century global workforce. Though group work is often challenging, it is a reality in nearly every employment setting. Learning to be a supportive team member, resolve conflicts, and discover your role preferences within group projects is an important part of the skills you will develop through your FAU education.

1. The team should be of up to 3 students. Finding team partners is primarily the students' responsibility. Choose your partners carefully, identify if your goals in this course are aligned, and if the level of commitment is the same. If there are differences between these two basic criteria, chances are you will not collaborate effectively, and there will be problems down the road. It is to your best advantage to document (email) your communications to avoid complications; if you feel more comfortable, use Google Docs to complete your project and/or feel free to cc: your emails to me. If there is an uneven number of

students in the class, the instructor will have the discretion to place the remaining student in a team whose team members should do everything possible to work together as a team.

2. While you will collaborate with your peers on the team projects, each team member must contribute an equal portion of effort or work towards each phase of the project. Based on observations of group work by the professor, team members contributing more than their fair share may receive higher grades than other group members. It is the group's responsibility to advise the professor of any issues or challenges that may hinder your ability to complete any phase of the project.

This project is broken down into tasks, in order to facilitate the success of your group project:

1. Project Deliverable 1: Team Charter (up to 10 points)
   You will connect with your group and create a Team Charter.
2. Project Deliverable 2: Data and Business Question (up to 20 points)
   The instructor will share a data file with the class. Each group carefully review the data to understand the content. Each group needs to propose a business question to answer.
3. Project Deliverable 3: Data Analytics Final Project (up to 100 points)
   Your group will perform detailed analyses on the data based on the chosen business question and present a full report of your analyses and findings of your project.

Detailed instructions, helpful resources, and grading rubrics are provided in the Canvas course. Most assignment pages will close one week after the stated deadline and time.

The instructor will calculate your grade based on the following weighted distribution:

Weighted Grade Category / Assignments / Points	Category = % total grade
Assignments	50%
Discussion	10%
Final Project	40%

TOTAL:	100%

Code of Academic Integrity

Students at Florida Atlantic University are expected to maintain the highest ethical standards. Academic dishonesty is considered a serious breach of these ethical standards, because it interferes with the university mission to provide a high quality education in which no student enjoys an unfair advantage over any other. Academic dishonesty is also destructive of the university community, which is grounded in a system of mutual trust and places high value on personal integrity and individual responsibility. Harsh penalties are associated with academic dishonesty. For more information, see University Regulation 4.001.

Attendance Policy Statement

Students are expected to attend all their scheduled University classes and to satisfy all academic objectives as outlined by the instructor. The effect of absences upon grades is determined by the instructor, and the University reserves the right to deal at any time with individual cases of non-attendance. Students are responsible for arranging to make up work missed because of legitimate class absence, such as illness, family emergencies, military obligation, court-imposed legal obligations, or participation in University-approved activities. Examples of University-approved reasons for absences include participating on an athletic or scholastic team, musical and theatrical performances, and debate activities. It is the student's responsibility to give the instructor notice prior to any anticipated absences and within a reasonable amount of time after an unanticipated absence, ordinarily by the next scheduled class meeting. Instructors must allow each student who is absent for a University-approved reason the opportunity to make up work missed without any reduction in the student's final course grade as a direct result of such absence.

The class teaching mode is "In-person with live remote option", so it is expected that all students attend the either in-person or live remotely via Zoom. We might have graded in-class assignments to keep everyone engaged with the material.

Religious Accommodation Policy Statement

In accordance with the rules of the Florida Board of Education and Florida law, students have the right to reasonable accommodations from the University in order to observe religious practices and beliefs regarding admissions, registration, class attendance, and the scheduling of examinations and work assignments. University Regulation 2.007, Religious Observances, sets forth this policy for FAU and may be accessed on the FAU website at www.fau.edu/regulations.

Any student who feels aggrieved regarding religious accommodations may present a grievance to the executive director of The Office of Civil Rights and Title IX. Any such grievances will follow Florida Atlantic University's established grievance procedure regarding alleged discrimination.

Time Commitment Per Credit Hour

For traditionally delivered courses, not less than one (1) hour of classroom or direct faculty instruction each week for fifteen (15) weeks per Fall or Spring semester, and a minimum of two (2) hours of out-of-class student work for each credit hour. Equivalent time and effort are required for Summer Semesters, which usually have a shortened timeframe. Fully Online courses, hybrid, shortened, intensive format courses, and other non-traditional modes of delivery will demonstrate equivalent time and effort.

Course Grading Scale

Letter Grade	Percentage
A	100 - 93%
A-	< 93 - 90%
B+	< 90 - 87%
B	< 87 - 83%
B-	< 83 - 80%
C+	< 80 - 77%
C	< 77 - 73%
C-	< 73 - 70%
D+	< 70 - 67%
D	< 67 - 63%
D-	< 63 - 60%

Letter Grade	Percentage
F	< 60 - 0%

Grade Appeal Process

You may request a review of the final course grade when you believe that one of the following conditions apply:

- There was a computational or recording error in the grading.
- The grading process used non-academic criteria.
- There was a gross violation of the instructor's own grading system.

University Regulation 4.002 of the University Regulations contains information on the grade appeals process

Policy on Make-up Tests, Late work, and Incompletes

Please reach out to your instructor as early as possible and let him know if you cannot finish a task on time and provide a reason. If approved, you will be allowed to have a late submissions. But in general:

Assignments and Project deliverables: Late submissions will not be accepted.

Online Discussions: Late submissions will not be accepted.

Policy on the Recording of Lectures

Students enrolled in this course may record video or audio of class lectures for their own personal educational use. A class lecture is defined as a formal or methodical oral presentation as part of a university course intended to present information or teach students about a particular subject. Recording class activities other than class lectures, including but not limited to student presentations (whether individually or as part of a group), class discussion (except when

incidental to and incorporated within a class lecture), labs, clinical presentations such as patient history, academic exercises involving student participation, test or examination administrations, field trips, and private conversations between students in the class or between a student and the lecturer, is prohibited. Recordings may not be used as a substitute for class participation or class attendance and may not be published or shared without the written consent of the faculty member. Failure to adhere to these requirements may constitute a violation of the University's Student Code of Conduct and/or the Code of Academic Integrity.

Artificial Intelligence Preamble

FAU recognizes the value of generative AI in facilitating learning. However, output generated by artificial intelligence (AI), such as written words, computations, code, artwork, images, music, etc., for example, is drawn from previously published materials and is not your own original work.

FAU students are not permitted to use AI for any course work unless explicitly allowed to do so by the instructor of the class for a specific assignment. **[Policy 10.16 Artificial Intelligence]**

Class policies related to AI use are decided by the individual faculty. Some faculty may permit the use of AI in some assignments but not others, and some faculty may prohibit the use of AI in their course entirely. In the case that an instructor permits the use of AI for some assignments, the assignment instructions will indicate when and how the use of AI is permitted in that specific assignment. It is the student's responsibility to comply with the instructor's expectations for each assignment in each course. When AI is authorized, the student is also responsible and accountable for the content of the work. AI may generate inaccurate, false, or exaggerated information. Users should approach any generated content with skepticism and review any information generated by AI before using generated content as-is.

If you are unclear about whether or not the use of AI is permitted, ask your instructor before starting the assignment.

Failure to comply with the requirements related to the use of AI may constitute a violation of the **Florida Atlantic Code of Academic Integrity, Regulation 4.001.**

Proper Citation: If the use of AI is permitted for a specific assignment, then use of the AI tool must be properly documented and cited. For more information on how to properly cite the use of AI tools, visit **https://fau.edu/ai/citation**

AI Language Specific To This Course

AI Flexible: The use of AI to assist in work assigned in this specific course is permitted only for specific assignments as indicated by the instructor. Use must be properly documented and cited per instructor guidelines (https://fau.edu/ai/citation).

Counseling and Psychological Services (CAPS) Center

Life as a university student can be challenging physically, mentally and emotionally. Students who find stress negatively affecting their ability to achieve academic or personal goals may wish to consider utilizing FAU's Counseling and Psychological Services (CAPS) Center. CAPS provides FAU students a range of services – individual therapy, group therapy, and crisis services, to name a few - offered to help improve and maintain emotional well-being. For more information, go to http://www.fau.edu/counseling/

Student Support Services and Online Resources

- Center for Learning and Student Success (CLASS)
- Counseling and Psychological Services (CAPS)
- FAU Libraries
- Math Learning Center
- Office of Information Technology Helpdesk
- Center for Global Engagement
- Office of Undergraduate Research and Inquiry (OURI)
- Science Learning Center
- Speaking Center
- Student Accessibility Services
- Student Athlete Success Center (SASC)

- Testing and Certification
- Test Preparation
- University Academic Advising Services
- University Center for Excellence in Writing (UCEW)
- Writing Across the Curriculum (WAC)

Course Topical Outline

Some of the modules may be updated during the semester.

Module 1: Basics of Python Programming Part 1	Module 2: Basics of Python Programming Part 2
Module 3: Importing Data, Part 1	Module 4: Importing Data, Part 2
Module 5: Data Analysis with Python, Part 1	Module 6: Data Analysis with Python, Part 2
Module 7: Data Analysis with Python, Part 3	Module 8: Reviewing Some of the Concepts in Data Analytics
Module 9: Basics of R Programming, Part 1	Module 10: Basics of R Programming Part 2
Module 11: Data Visualization with R	Module 12: Data Analysis with R, Part 1
Module 13: Data Analysis with R, Part 2	

Title IX Statement

In any case involving allegations of sexual misconduct, you are encouraged to report the matter to the University Title IX Coordinator in the Office of Civil Rights and Title IX (OCR9). If University faculty become aware of an allegation of sexual misconduct, they are expected to report it to OCR9. If a report is made, someone from OCR9 and/or Campus Victim Services will contact you to make you aware of available resources including support services, supportive measures, and the University's grievance procedures. More information, including contact information for OCR9, is available at **https://www.fau.edu/ocr9/title-ix/**. You may also contact Victim Services at **victimservices@fau.edu** or 561-297-0500 (ask to speak to an Advocate) or schedule an appointment with a counselor at Counseling and Psychological Services (CAPS) by calling 561-297-CAPS.